SIMATS ENGINEERING
ASSESSMENT TOOL 2:

Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Total Marks: 30

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Code of Conduct

I L:Sathyam Sharma (192526028 2) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L Sath

Assessment Tasks

Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
   Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

L-Sathyam Sharma
1925260282
CSA1519
CLOUD COMPUTING AND
DATA ANALYSIS

CASE STUDY 1: Small Retail Inventory Management system

AIM:-
To study how cloud computing is used to manage retail inventory monitor products sales and forecast customer demanded

Introduction:-
It is a cloud based application that helps retails stores manage product stock and sales difficulty. store devices, such as point of sale (POS) systems, communicate with centralized cloud services through API

Architecture and components:-
clients:- POS systems, computers, and store dashboards.

API :- Transfer sales and inventory information btwn cloud services

Platform service :- Platform data
demand Forecasting

security :- VPNs, Firewalls, API gateways and access controll protects the system

WORKING Principle :-

. A customer purchases a product at a retail store

. The POS system records the transaction

. The cloud database updates the variable cost.

. Analytics service examine sales patterns.

. Demand Forecasting estimates Future Product requirements.

Advantages :-

- Real time monitoring
- Reduced stock shortages and overstocking
- improved sales analysis
- centralized management of multiple stores
- Better centralized Forecasting.
- secure access to inventory information

challenges :-

Dependence on Network connectivity

Risk of unauthorized access to buisness data

cloud services and storage costs

Incorrect Forecasting if sales data is incomplete.

RESULT :-

cloud computing enables the retail chain to manage inventory centrally analyze, sales data, make better purchasing deussions

CASE-2

CLOUD - Based Document collaboration System

AIM :-

TO Study how cloud computing supports online documents creation, real-time collaboration, and secure file sharing

INTRODUCTION :-

users access application such as google docs through web browsers. Documents are stored in cloud storage

version controls maintains previous document versions.

ARCHITECTURE AND COMPONENTS :-

clients :- web browsers; laptops, tablets, mobile phones.

web Application : Provides the document earlier and collaboration interface

web APIs :- Edits and exchange between clients and servers

cloud storage : stores documents and related files

version control : maintain previous version and supports recovery of Earlier edits

security :- Encryption, identity, management, authentication, and access permission Protect documents

ORKING PRINCIPLE :-

. A user sign in to the document collaboration platform
. The user creates a documents or open an existing one
. The document is retrieved from cloud storage
. The user edits the documents in the browser
. changes are transmitted to the backend through the APIs
. other authorized users receive synchorized updates
. The system saves documents changes and maintain version history

ADVANTAGES :-

Real time collaboration among multiple users
Access from different devices and locations
Easy documents sharing
version history and recovery
Reduced need to install desktop softwa

CHALLENGES

- Internet connectivity is required for full collaboration
- unauthorized sharing can expose confidential documents
- synchornization conflict may occur
- High user traffic can affect performance

RESULT :-

cloud computing enables users to create, edit and share documents collaboratively, while centralized storage and version control improves accessibility and reliability

ONLINE FOOD ODERING PLATFORM :

AIM :-

To study how cloud computing helps an online foods ordering platform through web application, RESTFUL APIs cloud storage, and scalable infrastructu

INTRODUCTION :-

Customers uses websites or mobile apps to order food RESTFUL APIS connect to the application backend services. cloud storage food images and invoices, while virtual machines and Kubernetes supports scalability

CASE-3

ONLINE FOOD ORDERING PLATFORM

AIM:

TO study cloud computing in online food ordering

INTRODUCTION :-

customers order food through websites or mobiles apps using cloud services

WORKING PRINCIPLE :-

- customers browse restaurant menus using a website or mobile app
- They select food items and place an order
- Restful web APIs send order details to backend cloud services

- Restaurants recieve and prepare orders

ADVANTAGES :-

Easy Food ordering through websites and mobile Apps

Access to menus at anytime and anywhere

Secure storage of customers and order details

scalable and cost effective infrastructure

CONCLUSION :-

The online Food ordering platform demonstrates how cloud computing integrates applications, APIs, storage, and infrastructure to provide secure efficient and scalable food ordering services